

Project Initialization and Planning Phase

Project Proposal (Proposed Solution) report

Date	3 July 2026
Team ID	
Project Title	India's Agriculture Crop Production Analysis
Maximum Marks	5 Marks

PROJECT OVERVIEW

Objective	To analyze and visualize crop production trends across India using multi-source agricultural data, enabling data-driven insights for better decision making by farmers, researchers, policymakers, and stakeholders.
Scope	Includes collection and integration of crop production data by type, state, season, and year; data cleaning and preprocessing; exploratory data analysis; trend forecasting; and interactive dashboards for insights and comparisons.

PROBLEM STATEMENT

Description	India's crop production data is scattered across multiple sources in inconsistent formats, making it difficult to analyze trends, compare performance across states and seasons, and predict future production accurately.
Impact	This leads to delayed insights, inefficient resource planning, improper policy formulation, and limited access to actionable information for farmers, researchers, and government agencies, affecting productivity and food security.

PROPOSED SOLUTION

Approach	Develop a data analytics platform that aggregates, cleans, and analyzes crop production data using statistical analysis, machine learning forecasting, and interactive visualization tools to deliver actionable insights.
Key Features	- Comprehensive crop production database by state, district, season, and year. - Interactive dashboards and visualizations (charts, maps, trend lines) for easy exploration. - Trend analysis and forecasting models for future crop production. - Comparative analysis across crops, regions, and time periods. - Exportable reports and data summaries for stakeholders.

INDIA'S AGRICULTURE CROP PRODUCTION ANALYSIS

Empowering Farmers. Enriching Agriculture. Ensuring Food Security.

SKILL WALLET

A SmartBridge Product

RESOURCE REQUIREMENTS

RESOURCE TYPE	DESCRIPTION	SPECIFICATION/ALLOCATION
 Hardware		
Computing Resources	CPU/GPU specifications, number of cores	NVIDIA T4 / A10G GPU Instances (AWS / GCP)
Memory	RAM specifications	32 GB High-Speed System RAM
Storage	Disk space for data, models, and logs	1 TB NVMe SSD Cloud Storage
Network	Bandwidth for data transfer and API calls	High-Speed Internet (Min. 100 Mbps)
 Software		
Frameworks	Programming / Web frameworks	Python (Backend), FastAPI, React.js (Frontend)
Libraries	Data processing, ML, and analytics libraries	pandas, numpy, scikit-learn, TensorFlow, GeoPandas, Prophet, XGBoost
Development Environment	IDE / Notebook Environment	VS Code, Jupyter Notebook
Tools & Platforms	Cloud services and DevOps tools	Docker, GitHub, Postman, AWS / GCP / Azure Services
 Data		
Data Source 1	Source, size, format	Govt Databases: data.gov.in, DAC&FW, IMD Weather API (~10+ GB, CSV/JSON)
Data Source 2	Source, size, format	Satellite Imagery: Sentinel-2, MODIS (NASA) (~50+ GB, GeoTIFF)
Data Source 3	Source, size, format	Soil Health & Crop Statistics (NBSS&LUP, ICAR) (~5+ GB, CSV/Excel)